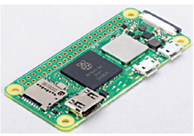
u-blox's expertise in chipset and module design into a compact and powerful component. The SiP supports Bluetooth mesh, long range and direction finding, Thread and Zigbee, operation up to 105°C, and is globally certified.

u-blox

www.u-blox.com

Upgraded Raspberry Pi board

The Raspberry Pi Zero 2 W offers enhanced features for a range of smart home applications and other IoT projects—all in a 65mm × 30mm small form factor. Built around the Broadcom BCM2710A1 SoC with 512MB of LPDDR2 SDRAM, it incorporates a quad-core 64-bit Arm Cortex-A53 CPU, clocked at 1GHz. At the heart of the board is a Raspberry Pi RP3A0 system-in-package (SiP). The upgraded processor provides Raspberry Pi Zero 2 W with 40% more single-threaded performance and 5x more multi-threaded performance than the original single-core Raspberry Pi Zero. An in-built wireless LAN with improved RF compliance gives more flexibility when designing with it.



Raspberry Pi

www.raspberrypi.com

Dual-output DC-DC converters

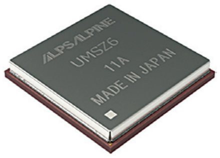
The PKE532x series is a new 30W dual output variant of board-mounted DC-DC converters. It has an 18-75V DC (100V peak) input range suitable for 24, 28, or 48V nominals. The PKE332x series input range has 9-36V DC (50V peak). Both parts are available with either ±12V or ±15V fully regulated dual outputs, with power ratings of 30W total. A particular advantage of these new variants is their ability to operate to over 100°C ambient with airflow and derating. Input-output isolation of the DC-DC converters is 1,500V DC, meeting basic insulation and safety requirements according to IEC/UL 62368-1.

Flex Power Modules

www.flexpowermodules.com

High-accuracy GNSS module

The UMSZ6 series GNSS module realises high-accuracy positioning within 50cm without correction data, a world-first for automotive applications. Even on general roads (approx. three metres wide), the module reliably enables vehicle positioning down to the lane level, as is required of various V2X applications, thereby contributing to greater sophistication of autonomous driving functions. Under joint development with Furuno, the multi-frequency GNSS receiver chip is based on Furuno's Extended Carrier Aiding technology. Running costs associated with RTK4 base stations, correction data receiving, and correction data use are no longer needed, maximising cost performance.



Alps Alpine

www.alpsalpine.com

Embedded graphics modules

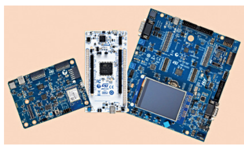
The new embedded MXM graphics modules are based on the NVIDIA Ampere architecture and offer up to 5,120 CUDA cores, 160 Tensor cores, and 40 RT cores with support for PCIe Gen 4 and up to 16GB GDDR6 memory at up to 115 watts of TGP. The new embedded graphics modules deliver real-time ray tracing, AI-accelerated graphics, and energy-efficient AI inference acceleration in the compact mobile PCI express (MXM) form factor. These modules enhance responsiveness, precision, and reliability for mission-critical, time-sensitive applications in healthcare, manufacturing, transportation, gaming, and other sectors.

ADLINK Technology

www.adlinktech.com

Energy-efficient evaluation boards

Based on the latest STM32U5 microcontrollers (MCUs), STMicroelectronics has announced new STM32Cube software packs and tools, as well as evaluation boards to accelerate development. The STM32U575 and STM32U585 MCUs combine security and power-saving innovations with the increased performance and energy efficiency of the latest Arm Cortex-M33 core. With up to 2MB of Flash memory on-chip, typical applications include smart, connected consumer products such as activity trackers and smart watches, as well as smart-home equipment, utility meters, equipment for industrial sensing and signaling, mobile point-of-sale terminals, insulin pumps, and glucose meters.



STMicroelectronics

www.st.som

DC-DC buck converters

The through-hole, 1/32nd brick-footprint, non-isolated DC-DC buck converters—the RPMA-4.5 and RPMA-8.0 with 4.5A and 8A output ratings, respectively—have a wide 9 to 53V input and trimmable, fully regulated